Distinct yet coupled hemispheric codes for foveal visual processing

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Conflict of interest statement

The authors declare no competing financial interests.

Acknowledgements

This work was supported by Australian Research Council (ARC) Discovery Early Career Researcher Awards awarded to A.K.R. (DE200101159) and TG (DE230100380), ARC Discovery Projects awarded to T.A.C. (DP160101300 and DP200101787) and National Institute of Health grants (RO1EY027018 and EY026701) to MB. MB also acknowledges support from P30 CORE award EY08098 from the National Institute of Health, and unrestricted supporting funds from The Research to Prevent Blindness Inc, NY, and the Eye & Ear Foundation of Pittsburgh. We would like to acknowledge the University of Sydney HPC service for providing High Performance Computing resources.

Abstract

Seeing the world feels effortless, yet it is accomplished by two hemispheres that do not process visual information in the same way. Whether this involves distinct processing cascades, tight hemispheric coupling, or something in between has fundamental implications for understanding the emergence of coherent perception from divided processes. Images presented to the fovea project to both hemispheres, offering an opportunity to isolate intrinsic hemispheric differences in visual coding. Here, we used electroencephalography and neural decoding methods to investigate the dynamics of foveal visual processing in the left and right hemispheres. Human participants (*N* = 20; 15 females, 5 males) viewed images of objects, faces and words in rapid sequences while performing an orthogonal task. We found different trajectories of visual coding within the left and right hemispheres, and these differences were characterised by distinct featural biases in each hemisphere, with a particular left bias for rectilinearity and right bias for colour. Yet, despite encoding visual information differently, the left and right hemispheres appear to converge on a shared representation. The results provide new insights into hemispheric dynamics underlying visual perception and the complementary roles of the left and right hemispheres in processing visual information.

Significance statement

The human brain is divided into two hemispheres that must work together to produce seamless visual perception. Despite well-documented asymmetries, what each hemisphere distinctly contributes to visual processing has remained unclear. Here, we used time-resolved neural recordings and pattern analysis methods to show that the left and right hemispheres have distinct and shared visual signatures: the left is preferentially sensitive to angular features, while the right is biased toward colour and coarse spatial structure. Despite these differences, the two hemispheres remain tightly coordinated in time. These findings reveal that the hemispheres contain intrinsic low-level visual feature sensitivities, yet their coordination allows the divided brain to construct a unified visual experience.

Introduction

The human brain is a divided but highly integrated system, comprising two hemispheres that must coordinate for seamless perception. The left and right hemispheres are largely symmetric in functional connectivity (Raemaekers et al., 2018) and have notable homotopic coupling between matched regions (Bryant et al., 2025), yet are asymmetric for many functions (Ocklenburg and Guo, 2024). The bilateral architecture likely has computational advantages, where redundancy affords robustness and flexible decision-making by distributing information across neural populations, whereas specialisation allows efficiency and division of labour (D'Alessandro et al., 2026; Liu et al., 2026). However, the division poses a fundamental challenge for perception, which must ultimately be unified. The two hemispheres seem to construct qualitatively different representations from the same input, with systematic asymmetries across multiple levels of processing (Davidoff, 1976; Hellige, 1996; Kanwisher et al., 1997; Cohen et al., 2000; Corballis, 2002; Stevens et al., 2012; Robinson et al., 2025). Characterising hemispheric differences and their integration is therefore necessary for understanding visual perception.

Behavioural research using lateralised presentation has revealed that the two hemispheres have different biases for the same input. The left hemisphere appears biased toward high spatial frequencies and local detail; the right toward low spatial frequencies and global structure (Sergent, 1982; Ivry and Robertson, 1998; Yovel et al., 2001). These asymmetries are not necessarily based at the earliest stages of processing: colour lateralisation varies with task demands, with right hemispheric bias for colour matching (Davidoff, 1976) but left bias for colour naming (Ting Siok et al., 2009). Similarly, spatial frequency asymmetries may partially reflect decision-level biases (Peterzell et al., 1989; Christman et al., 1991). Notably, this body of work has relied on peripheral stimulation, leaving open whether the same asymmetries govern foveal processing, where receptive fields, feature tuning, and cortical representation differ markedly (Van Essen et al., 1984; Wandell et al., 2007; Stewart et al., 2020).

At the level of visual categories, face processing tends to be right-lateralised (Kanwisher et al., 1997; Rossion et al., 2003) and word processing left-lateralised (Cohen et al., 2000; Dehaene et al., 2010). Scene processing also shows right hemispheric biases (Stevens et al., 2012; Zhen et al., 2017), consistent with visuospatial specialisation. Yet, even highly lateralised functions rely on bilateral processing (Behrmann and Plaut, 2014), and lateralisation itself has been proposed to arise via cross-hemispheric communication (Ocklenburg and Guo, 2024). Interhemispheric integration is therefore not incidental but fundamental to visual perception, as underscored by striking failures following callosal lesions (Gazzaniga, 2000). These category asymmetries raise the question of whether they reflect high-level identity and/or the low-level features that distinguish these stimuli. Answering this requires characterising the visual dimensions preferentially encoded by each hemisphere and the degree to which these representations are independent versus shared.

We previously used electroencephalography (EEG) with lateralised stimuli to show that interhemispheric transfer prioritises conceptual meaning over visual statistics (Robinson et al., 2025). Here, we focus on foveal stimulation, which results in a split representation where both hemispheres receive direct retinal input from the stimuli (Lavidor and Walsh, 2004). Precise hemispheric coordination may be especially critical here, as fine-grained foveal discrimination requires integration of high-fidelity input. Although behaviour cannot dissociate hemispheric contributions for foveal stimuli, the millisecond resolution of EEG allows assessment of hemispheric dynamics. Participants were presented with central images of objects, faces, and words, and using time-resolved decoding (Carlson et al., 2020; Robinson et al., 2023) and representational similarity analyses (RSA; Kriegeskorte et al., 2008), we characterised what information each hemisphere represents and when, and the degree to which representations were shared or distinct across hemispheres. We found feature-specific hemispheric biases: the left hemisphere preferentially coded rectilinear features, while the right hemisphere was biased toward colour content. Despite these differences, the hemispheres showed tightly coupled temporal representations and both hemispheres exhibited a coarse-to-fine processing cascade. Together, these results illuminate how the distinct but coordinated representations of each hemisphere contribute to unified visual perception for centrally viewed stimuli.

Materials and methods

The data reported here were collected as part of a larger experiment described in Robinson et al (2025), in which images were presented either lateralised to one hemifield or foveally. We previously reported findings focusing on hemispheric processing of lateralised images (Robinson et al., 2025), whereas here we focus on the representational dynamics of the central image condition, an important complement and means of comparison relative to the previous findings.

Code accessibility

All code will be made available on GitHub upon publication. Neural data are available on OpenNeuro (https://openneuro.org/datasets/ds005087).

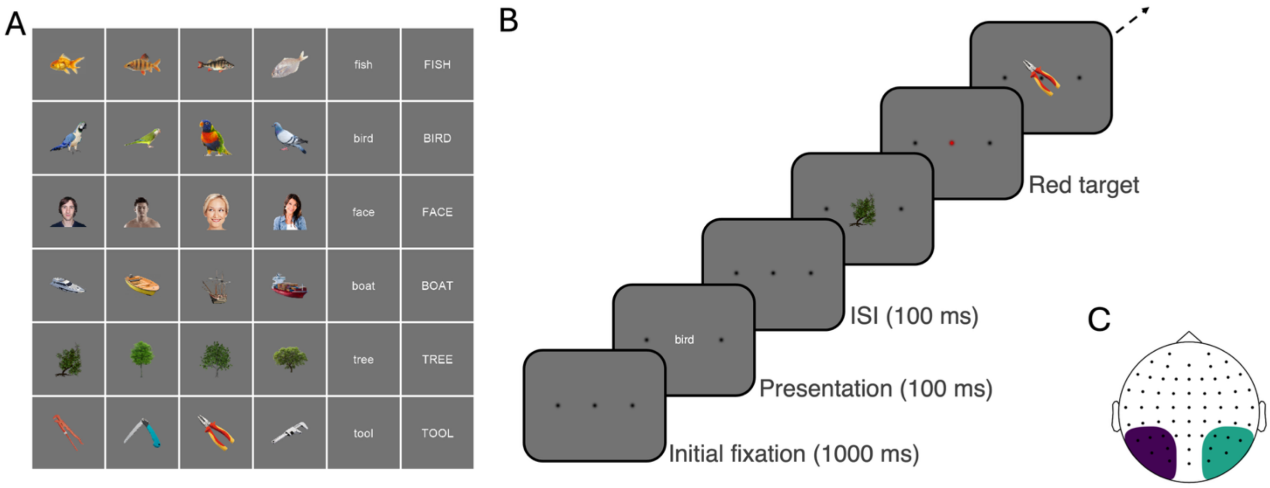
Participants

Participants were 20 adults recruited from the University of Sydney (15 females, 5 males; median age 22 years) in return for payment or course credit. The study was approved by the University of Sydney ethics committee and informed consent was obtained from all participants. Edinburgh handedness scores indicated that 17 participants were right-handed, 1 was left-handed and 2 were ambidextrous. Similar patterns of results were found when analyses were restricted to right-handed participants (see Supplementary Material).

Stimuli and Design

Participants viewed images that appeared on a 1920 x 1080 resolution ASUS gaming monitor while their neural activity was measured with EEG. There were 36 stimuli: four different image exemplars of 6 basic objects: fish, bird, face, boat, tree and tool; and word labels for the same objects in lower and upper case (Figure 1A). Half the trials were words and half were objects (i.e., each word exemplar was presented twice as often as the object stimuli to equate frequency of presentation). Images were presented centrally or to the left or right of fixation within a bounding box of 3.24 x 3.24 degrees of visual angle, but here we just focus on the centrally presented images (Figure 1B). All stimuli were presented using Psychopy (Peirce et al., 2019).

Figure 1. Experimental design.



Note. A) Stimuli were 36 images of real world objects and words from six different categories: fish, birds, faces, boats, trees and tools. B) Example trial timeline for the central image condition. Images were presented in rapid sequences for 100 ms duration with 100 ms inter-stimulus interval (ISI). Note the dots marking image positions and target are shown larger here for visualisation purposes. C) EEG electrodes for left hemisphere and right hemisphere clusters comprised six electrodes each.

Images were presented in sequences of 192 images (Figure 1B). Each stimulus was presented for 100ms with a gap of 100ms between each image (5 Hz trial presentation) to preclude the opportunity for saccades and loss of control of the visual field arrangement. We have shown previously that such rapid presentation designs afford sensitive within-subject power to assess independent image representations (Grootswagers et al., 2019, 2024; Robinson et al., 2019). There were 96 image stimuli per sequence (4 repeats per image) and 96 word stimuli (8 repeats per word stimulus). Within each sequence, images were presented in random order. There were 12 central sequences in total. Thus, throughout the experiment, there were 48 repeats of each of the 24 image stimuli and 96 repeats of each of the 12 word stimuli for the central condition. To maintain equal trial numbers per class, only 48 repeats for the word stimuli were analysed.

During the experimental session, participants were asked to perform an orthogonal task. They had to monitor the three dots marking the possible image locations (central fixation, left and right) and detect when one turned red (Figure 1B). They had to press a button when they detected a target. This task was designed to be irrelevant to the stimulus images, but, critically, ensured that participants maintained attention across the whole visual field throughout the experiment, regardless of image presentation condition. Performance on the task was high (mean accuracy = 87.45%, SD = 13.12%).

EEG recording and preprocessing

EEG data were continuously recorded from 64 electrodes arranged in the international 10–10 system for electrode placement (Oostenveld and Praamstra, 2001) using a BrainProducts ActiChamp system, digitized at a 1000Hz sample rate. Scalp electrodes were referenced to Cz during recording. The EEGLAB toolbox (Delorme and Makeig, 2004) was used to preprocess the data offline. Preprocessing steps are the same as in Robinson et al (2025); briefly, we interpolated noisy electrodes, applied a common average reference, filtered the data (highpass of 0.1Hz, lowpass of 100Hz), created epochs -100 to 800ms relative to stimulus onset, and performed a Laplacian transformation to enhance the local signal (Perrin et al., 1989).

Neural decoding

We investigated visual representational dynamics in each hemisphere. EEG data were analysed using time-resolved classification methods and implemented using the CoSMoMVPA toolbox (Oosterhof et al., 2016). Decoding was performed separately for left and right hemispheres using clusters of six occipito-temporal electrodes (Figure 1C). These electrodes were the same as in our previous work (Robinson et al., 2025) and mapped on to visually responsive regions as verified with searchlight analyses (see Supplementary Material). At each single time point, data were pooled across the 6 EEG sensors, and we tested the ability of a linear discriminant analysis (LDA) classifier to discriminate between the patterns of neural responses associated with 48 repeats of each stimulus. Decoding was performed for all pairs of image combinations for the 36 stimuli (e.g., face1 vs tree2, word-tool1 vs fish4), resulting in 630 unique contrasts across time.

Hemispheric dynamics were assessed by taking the mean decoding accuracy across all stimulus pairs for each time point per hemisphere, with chance level of 50%. Decoding analyses were performed separately for each participant, and the mean was then taken across the group. Accuracy of classifier predictions informed the information in the neural signal, where above-chance classification accuracy (>50%) indicated that the hemisphere contained information discriminating between images.

Image statistics

To characterise the low-level visual properties of our stimuli, we computed several image statistics for each stimulus. All stimulus images were first rendered by presenting them in PsychoPy using the exact parameters from the experiment (stimulus size of 128 pixels for images and height of 40 pixels for text, presented on a grey background) and capturing each frame to create 768 × 768 pixel images. All image statistics were computed on the rendered images including background, matching the visual input experienced by participants.

Luminance. Mean luminance was calculated by converting each RGB image to greyscale and computing the average pixel intensity across the entire image.

Colour. Colour information was quantified by converting images from RGB to CIE Lab colour space, which approximates human colour perception. We computed a joint 2D histogram over the a* (red-green) and b* (blue-yellow) chrominance channels, using 32 bins per channel (spanning −128 to 128), yielding a 32×32 histogram that captures the distribution of hues and saturations independent of luminance and without preserving spatial layout. The resulting values were vectorised to create a colour descriptor for each image.

Spatial frequency. We computed spatial frequency content using 2D fast Fourier transforms (FFT). Images were converted to greyscale, the mean pixel intensity was subtracted, then we applied a 2D FFT and shifted the zero-frequency component to the centre of the spectrum. We calculated a radial average profile by computing the mean FFT magnitude at each distance from the centre, collapsing across orientation to obtain a rotation-invariant measure of spatial frequency power. Given our stimulus parameters (237.04 pixels/degree, Nyquist frequency of 118.52 cycles/degree), each point in the 545-element radial profile corresponded to 0.22 cycles/degree. We derived separate measures for low spatial frequencies (0-3 cycles/degree, corresponding to the first 14 points of the radial profile excepting the DC offset) and high spatial frequencies (≥6 cycles/degree, corresponding to points 28 onwards).

Edge information. Locations of edges in each stimulus (e.g., object outline) were calculated using the edge() function in MATLAB on the greyscale images with the Canny method and a threshold of 0.1.

Rectilinearity. The degree of rectilinearity in each image was calculated using V-shaped Gabor filters adapted from the Rectilinearity Toolbox (Nasr et al., 2014). These filters are sensitive to corners and edges with specific vertex angles. The vertex angle (θ) was systematically varied to capture features ranging from straight edges (0°) to 30°, 60°,90°,120° and 150° to capture acute, right and obtuse corners.

Each image was first converted to greyscale and a Canny detector (Canny, 1986) with threshold of 0.1 was then applied to extract edge contours. V-shaped filters were generated for each combination of spatial frequency (1/24, 1/12 and 1/6 cycles/pixel) and vertex angle. Each filter was constructed by combining two Gaussian arms forming a V, and windowed with a Gaussian envelope to constrain its spatial extent. To detect angles at different orientations, each filter was applied at 16 orientations (0° to 337.5° in 22.5° steps) and responses were averaged.

For each rotation, the filter was convolved with the edge map, and the absolute value was raised to the power of 8 to enhance selectivity for strong matches. Responses were normalised (min-max) within each spatial frequency and angle across the stimulus set, then the mean was calculated to yield a single rectilinearity index per image.

Representational similarity analyses

To investigate the structure of representations in the two hemispheres, we used representational similarity analyses (RSA) (Kriegeskorte et al., 2008), which abstracts away from specific activity patterns (e.g., from strength of electrode responses over the left hemisphere) to the relationships between different representations. In this case, RSA allowed hemispheric-specific representations to be compared with representations of the other hemisphere (Robinson et al., 2025) as well as with models based on image features. To characterise neural and feature patterns, we constructed representational dissimilarity matrices (RDMs), which quantified the similarity between each image on each measure (neural, visual features). Each of these RDM models was a 36 x 36 matrix quantifying the dissimilarity for each of the 36 stimuli with each other stimulus. The RDMs were symmetrical across the diagonal, with 630 unique values.

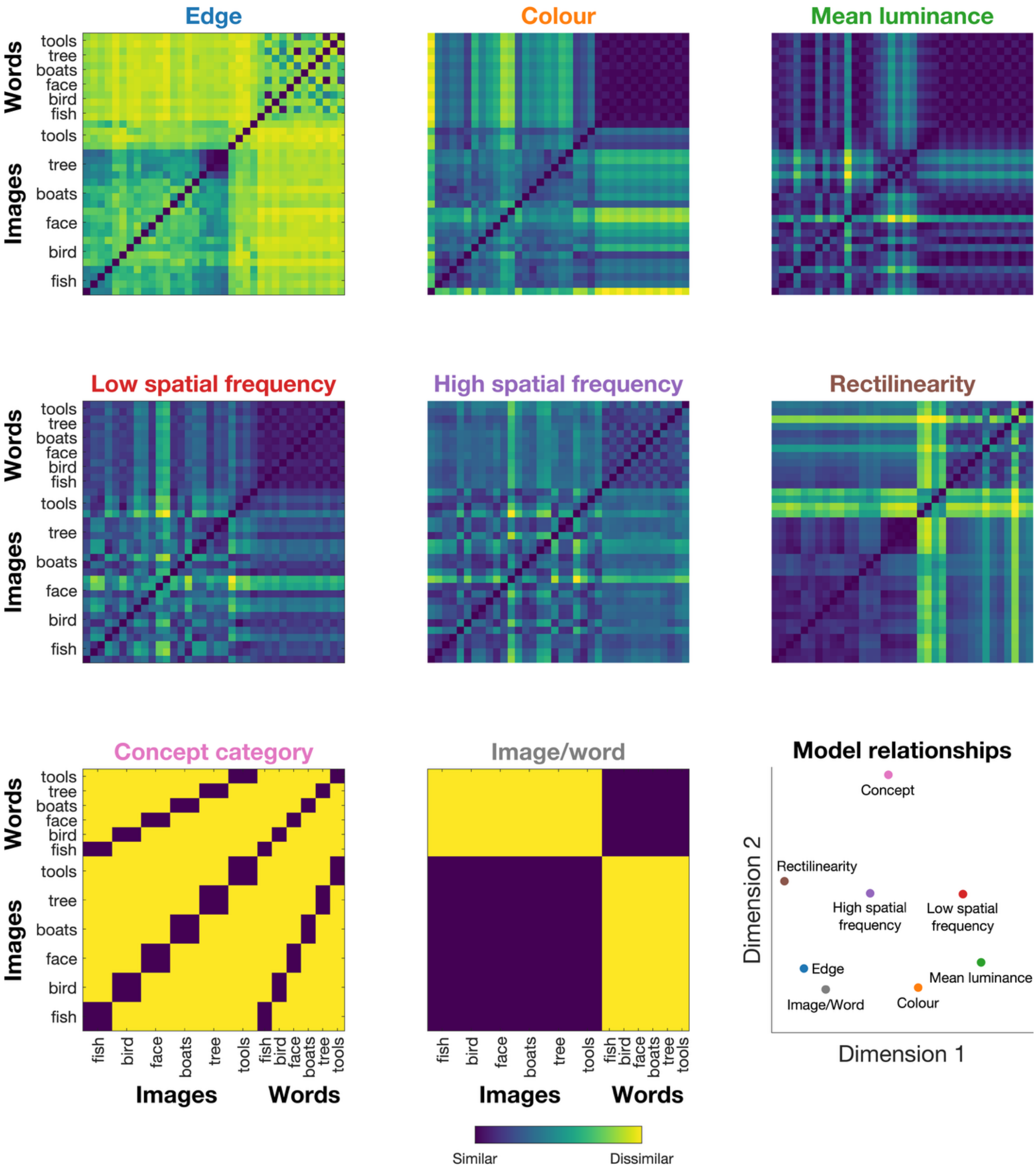
Neural RDMs were created for each hemisphere and time point per participant, and model RDMs were calculated for each of the visual statistics of interest. For the neural data, we constructed neural RDMs using decoding accuracy for each pair of images at each time point. Separate 36 x 36 stimulus RDMs were constructed for each hemisphere and time point, where each cell contained the mean decoding accuracy between two stimuli. For visual statistics, we constructed an RDM for each visual feature (e.g., rectilinearity) by computing pairwise Euclidean distance between all stimuli, except for edge content, for which we used Jaccard distance. This resulted in separate RDMs for edge content, colour, mean luminance, low spatial frequency, high spatial frequency and rectilinearity (Figure 2). As a comparison to these models, we also constructed categorical RDMs representing concept category membership and image versus word stimulus type. Spearman correlations revealed that the concept category model was significantly correlated with the rectilinearity model ($\rho = .10$, $p = .01$) but not other visual models, whereas the image/word model was significantly correlated with edge, colour, high spatial frequency and rectilinearity ($\rho > .31$, $p < .001$). Relationships between models are visualised in the last plot of Figure 2; see Supplementary Material for all correlations between models.

All RSA comparisons used Spearman correlation of the lower triangles of the RDMs (i.e., the 630 unique pairwise values). To assess the representations of image statistics in each hemisphere, we correlated each of the image statistics models with the neural RDMs per hemisphere over time.

For cross-hemispheric RSA, we investigated how representations varied across the hemispheres over time. This was done in a split-half manner, using decoding accuracy from odd and even sequences separately. To assess within-hemisphere dynamics, for each hemisphere, we correlated RDMs for each time point from odd sequences with each time point from even sequences, giving a measure of time-generalised within-hemisphere consistency. Then, we correlated the left and right hemisphere RDMs using odd sequences in the left hemisphere and even sequences in the right hemisphere, and vice versa, and taking the mean correlations at each pair of time points. This allowed us to assess similarity across versus within the hemispheres, and whether this similarity was dependent on transfer delays.

Spearman correlations were performed separately for each EEG participant, and the mean was taken across participants. Group mean correlations were tested statistically and reliable non-zero correlations were taken as evidence for the same geometric structure of object representations across the different models.

Figure 2. Visual feature relationships in the stimulus set.



Note. Dissimilarity matrices representing models of visual features, used to assess feature coding in each hemisphere. Six models were constructed based on visual statistics: edge, colour, luminance, low spatial frequency, high spatial frequency and rectilinearity. Two models were constructed based on canonical categories: concept, and image/word. The last plot shows multi-dimensional scaling of model relationships, where the distance between models indicates their similarity.

Experimental design and statistical analyses

This study used a within-subjects design, where all participants completed all conditions of the experiment. The primary independent variable was hemisphere (left, right) and the dependent variables were decoding accuracy and representational similarity correlations across time. Sample size is consistent with previous EEG decoding studies using similar rapid serial presentation paradigms (Grootswagers et al., 2019, 2024; Robinson et al., 2019)

We used Bayesian statistics to determine the evidence for the alternative relative to the null hypotheses (Jeffreys, 1961; Wagenmakers, 2007; Rouder et al., 2009; Dienes, 2011, 2016). For decoding analyses, the alternative hypotheses of above- chance (50%) decoding were tested. For correlation analyses, the alternative hypotheses of above- and below- zero correlations were tested. Bayes Factors do not require explicit correction for multiple comparisons in the traditional sense because they quantify the evidence in a continuous manner rather than producing binary decisions. Bayes Factors were calculated using the 'BayesFactor' package in R (Morey et al., 2018) using a JZS prior, centred around chance decoding of 50% (Rouder et al., 2009). The default scale factor was 0.707, meaning that for the alternative hypotheses of above- and below- chance decoding, we expected to see 50% of parameter values falling within -.707 and .707 standard deviations from chance (Jeffreys, 1961; Zellner and Siow, 1980; Rouder et al., 2009; Wetzels and Wagenmakers, 2012). Null intervals were specified as a range of effect sizes between -inf to 0.5 (decoding) or -0.5 to 0.5 (correlations)(Teichmann et al., 2022).

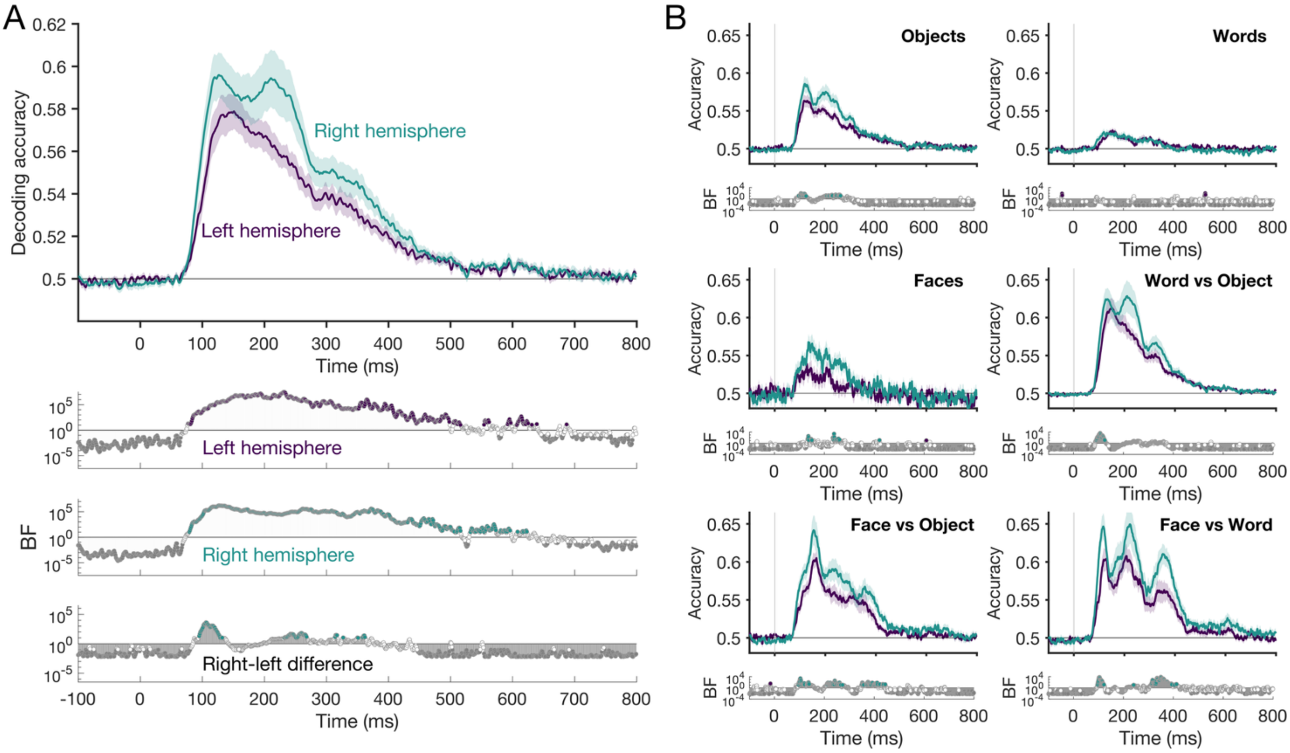
A Bayes factor (BF) is the probability of the data under the alternative hypothesis relative to the null hypothesis. In line with conventions, we consider BF > 3 as reliable evidence and BF > 10 for strong evidence for above-chance decoding or non-zero correlations. In line with our previous work (Robinson et al., 2025), we consider the "onset" to be the first period of sustained evidence (ten consecutive time points of BF>10). We interpret BF < 1/3 and BF < 1/10 as reliable and strong evidence in favour of the null hypothesis (Jeffreys, 1961; Wetzels et al., 2011).

Results

To characterise the time course of visual information processing in the brain, we first examined how the left and right hemispheres represented the visual stimuli. To do this, we assessed mean decoding accuracy across all 630 stimulus pairs, indexing stimulus discriminability per hemisphere (Figure 3A). Stimulus content was discriminable in both hemispheres, as indexed by above chance decoding accuracy, but the dynamics of information processing followed distinct patterns in the left and right hemispheres. For the left electrode cluster, decoding accuracy was reliably above chance continuously from 77ms to 498ms (BF_{10} > 10.58), with a peak at 151ms (57.89%, $BF_{10} = 3.72 \times 10^6$). For the right electrode cluster, decoding accuracy was reliably above chance continuously for a similar time period (79ms to 506ms; BF_{10} > 10.76), with a peak at 127ms (59.59%, $BF_{10} = 1.68 \times 10^6$) and a second clear peak at 211ms (59.44%, $BF_{10} = 5.27 \times 10^4$).

Next, to explore whether decoding performance and hemispheric trends were driven by stimulus category, we split the pairwise comparisons into six groups: within-face, within-word, within-object, face vs object, face vs word, and word vs object. Note that there were 24 objects, 4 faces and 12 words in the stimulus set, so each of these groups consisted of a different number of stimulus pairs. Hemispheric differences in decoding accuracy were assessed at each timepoint using Bayes factors. Overall, we found clear right hemispheric dominance, with stronger decoding accuracy in the right than the left hemisphere for all category subsets except within words (Figure 3B). Notably, the largest differences between the right and left hemispheres were for between-category contrasts (i.e., word vs object, face vs object, face vs word).

Figure 3. Distinct visual coding dynamics in each hemisphere

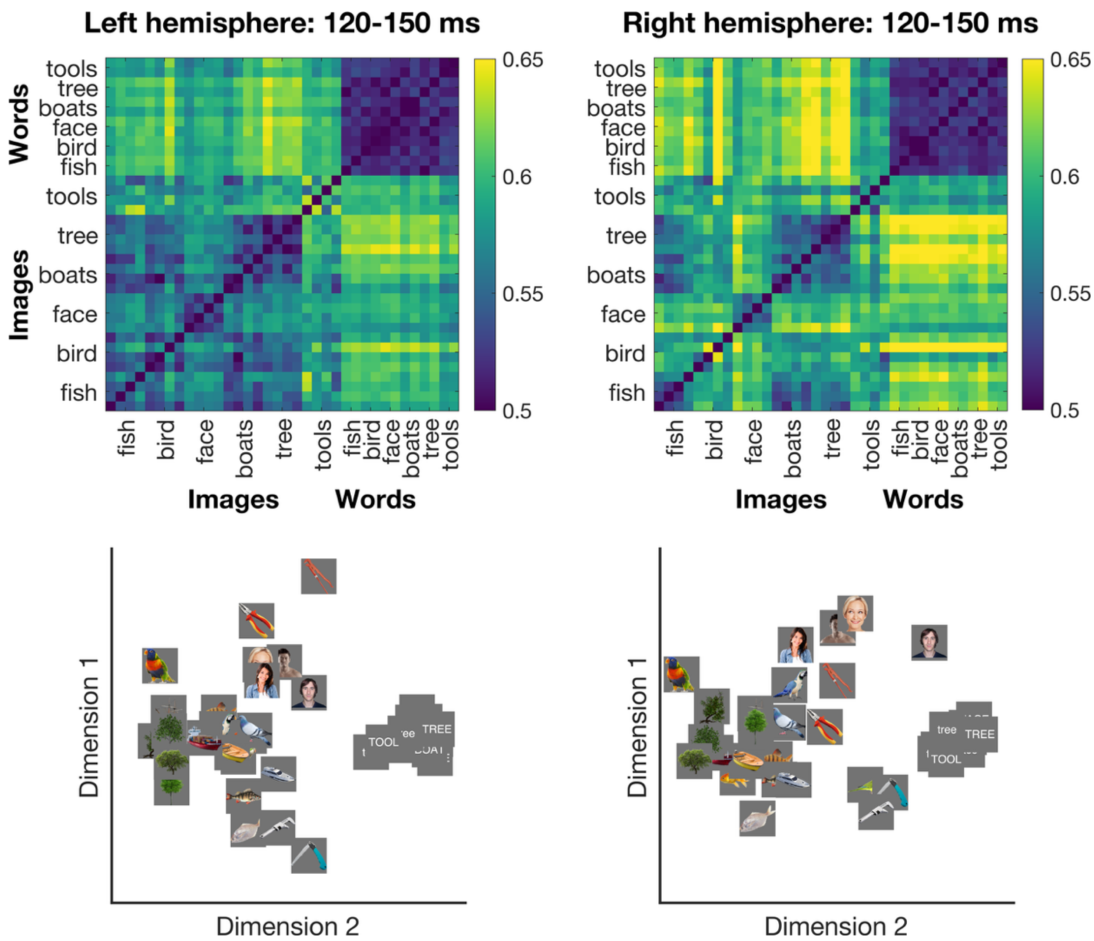


Note. Neural decoding reveals differences in visual representations in the left and right hemispheres. A) Mean decoding accuracy across all 630 pairs of experimental images. Bayes factors below quantify the evidence for above chance decoding per hemisphere and differences between hemispheres. Right-left difference is colour coded by the hemisphere with reliably stronger decoding. B) Mean decoding accuracy for different stimulus categories: within object, within face, within word, face versus object, word versus object and face versus word. Bayes Factors below each plot quantify the evidence for differences between the left and right hemisphere.

Hemispheric representational geometry

To assess how each hemisphere represents the visual stimuli, we used the pairwise decoding accuracy to construct models of representational geometry; that is, we assessed how each stimulus was coded relative to each other stimulus. Figure 4 shows neural RDMs and the two-dimensional visualisation of the similarities between stimuli in terms of their neural representations per hemisphere during a representative time window (120-150ms). Using RSA, we then compared hemispheric geometry with models of visual features, and compared geometry across the two hemispheres.

Figure 4. Neural dissimilarity across the stimulus set per hemisphere.



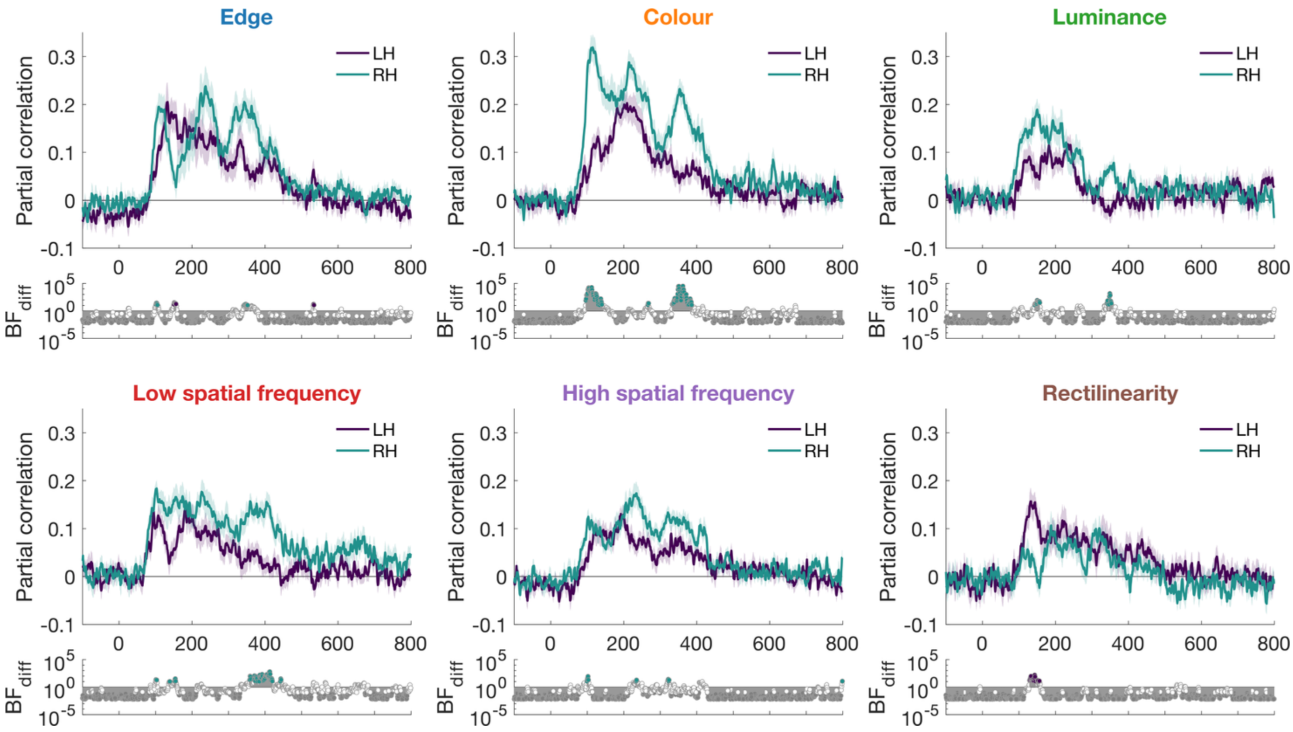
Note. Top: Group mean dissimilarity matrices for a time window 120-150 ms after stimulus onset, indexing decoding accuracy for the two hemispheres. Bottom: Multidimensional scaling for visualisation of top two dimensions accounting for relationships between stimuli. Images in the same category form distinct clusters, which is particularly evident for words.

Hemispheric feature biases

To characterise how each hemisphere represents visual features, we calculated correlations between time-resolved neural RDMs from the left and right hemispheres (like those in Figure 4) with a set of models based on image statistics of the stimuli (Figure 2) while partialling out the other hemisphere's RDM. This analysis revealed that each hemisphere contained unique feature-related representations over and above the other hemisphere. Both hemispheres had reliable partial correlations with all models (Figure 5).

However, the hemispheres diverged in their feature profiles (see Table 1 for onset times and peak correlations per feature and hemisphere). Low spatial frequency information was the earliest feature reflected in both hemispheres, consistent with rapid low-level feature extraction in a coarse-to-fine grading. All visual features, except for rectilinearity, had earlier onsets and stronger correlations in the right than left hemisphere. Rectilinearity was the only model to show a purely left hemispheric bias, with stronger and earlier correlations in the left than right hemisphere between 129-156 ms (BFs = 2.73-154.68, median = 13.93). In contrast, the right hemisphere showed stronger correlations than the left hemisphere for colour (96-140 ms, BFs = 6.10-1.86 x 10^4, median = 720.20), low spatial frequency information (152-155 ms, BFs > 11.60) and luminance (143-157 ms, BFs > 11.54). Edge information was initially stronger in the right hemisphere from 101-105ms (BFs>11.93), but subsequently stronger in the left hemisphere from 150-156ms (BFs>15.84). These effects were not driven by specific asymmetric categories; we ran the same analyses but restricted to the 20 objects in the stimulus set (i.e., no faces and words, which have documented right and left biases, respectively; see Supplementary Material). Though the correlations were noisier, which was expected due to less stimulus variation and fewer stimulus contrasts to correlate, the trend in the dynamics was similar: specifically, in comparison to the left hemisphere, correlations were numerically larger for colour and smaller for rectilinearity in the right hemisphere.

Figure 5. Feature representations per hemisphere



Note. Hemispheric-specific representations reflect image statistics of presented stimuli. Partial neural-feature correlations were higher in the right hemisphere than the left for all features except rectilinearity, which showed stronger correlations in the left hemisphere. Bayes Factor plots reflect differences in partial correlations between the left and right hemispheres.

Table 1. Timing of neural-feature correlation onset and peaks.

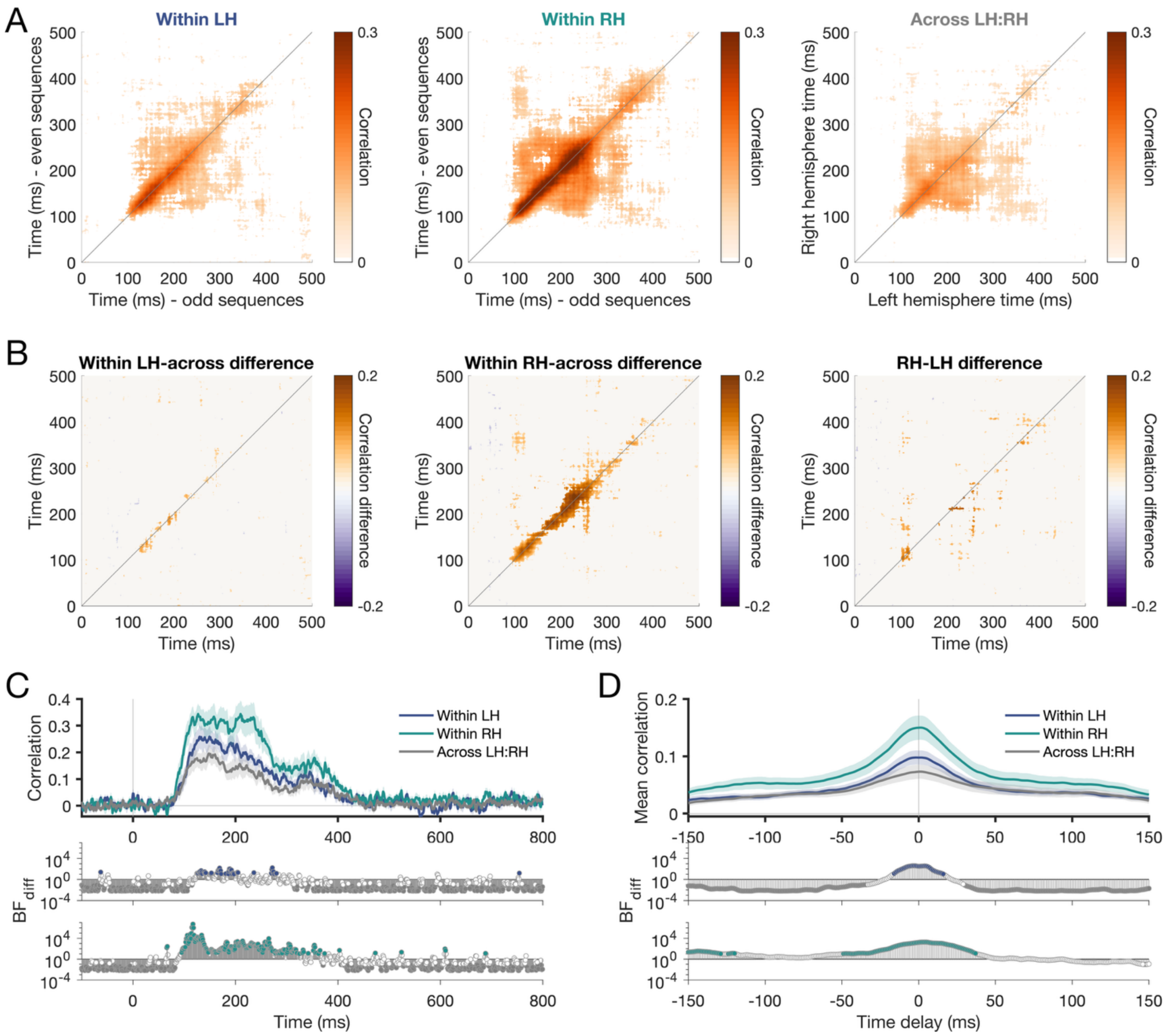
	Left hemisphere			Right hemisphere		
	Onset time	Peak time	Peak correlation	Onset time	Peak time	Peak correlation
Edge	102 [100 102]	**133 [131 179]**	0.21	**92 [90 92]**	237 [236 237]	0.24
Colour	110 [106 111]	209 [182 228]	0.20	**77 [76 84]**	**115 [111 117]**	0.32
Luminance	112 [111 209]	234 [182 241]	0.12	**95 [87 95]**	**150 [149 150]**	0.19
Low SF	89 [88 90]	181 [94 182]	0.14	**76 [76 79]**	103 [101 228]	0.18
High SF	105 [104 132]	**192 [192 199]**	0.13	**85 [82 87]**	235 [217 235]	0.17
Rectilinearity	**113 [113 114]**	**133 [132 144]**	0.16	299 [187 317]	188 [188 320]	0.11

Note. Onset time was calculated as the first of ten consecutive time points with reliable correlations (BF>10). Brackets denote 95% confidence intervals from leave-two-out jack-knifing (190 permutations). Bold values reflect the hemisphere with the reliably earlier time for that visual feature and decoding aspect (onset/peak).

Unique and shared hemispheric representations

Having established that the hemispheres differ in their feature profiles, we next determined the extent to which representational information is unique to each hemisphere versus shared in common across hemispheres. To address this, we performed a time-generalisation RSA in which neural RDMs were correlated in a split-half manner across all pairs of time points. Within-hemisphere correlations (left–left and right–right) index representational consistency, while across-hemisphere correlations (left–right) index interhemispheric information sharing. Within-hemisphere consistency was higher than across-hemisphere correlations, particularly along the diagonal where similar time periods were compared, indicating that each hemisphere contains representational structure not fully captured by the other (Figure 6). Examining the across-hemisphere correlations as a function of time delay revealed that interhemispheric similarity was greatest around zero lag, suggesting that information coding in the hemispheres is tightly aligned rather than reflecting systematic asymmetric transfer from one hemisphere to the other.

Figure 6. Unique information is contained within each hemisphere.



Note. Time generalisation RSA results: RDMs were correlated in a split-half manner for every pair of time points. Matrices show strength of correlation for reliable correlations only (BF>10). A) Within left hemisphere (LH) and right hemisphere (RH) correlations reveal *consistency* whereas across LH:RH correlations reveal interhemispheric information sharing. B) Within hemispheric consistency was higher than LH:RH interhemispheric sharing for both the LH and RH, particularly for tightly coupled time windows close to the diagonal (i.e., with short time delays). RH consistency was reliably higher than LH. C) Within versus across hemispheric consistency as a function of processing time (time delay = 0 ms; same as diagonals in A). Higher within than across correlations suggest unique information in each hemisphere relative to the information shared across hemispheres. D) Mean correlation as a function of time delay. Greatest correlations were observed for short delays. Within LH and RH were reliably higher than Across LH:RH correlations, primarily for short time delays (<50ms). Bayes Factor plots underneath reflect Within-Across differences per hemisphere.

Discussion

In this study, we investigated the temporal dynamics of visual representations within, and between, the left and right hemispheres in response to visual input presented foveally and relayed to both hemispheres. Using EEG and neural decoding methods, we characterised how object, face, and word representations unfold over time in each hemisphere, what visual features drive these representations, and the extent to which hemispheric representations are distinct versus shared. Our findings reveal that the two hemispheres carry complementary rather than redundant visual information, characterised by specific featural biases and different temporal dynamics, yet remain tightly temporally coupled.

Featural biases distinguish the hemispheres

The core finding of this study is that each hemisphere contains unique representational information that is not fully captured by the other. Here, the visual input was presented foveally, which means that the left and right hemispheres both received direct retinal input from each image, contralateral to the split at the vertical midline (Lavidor and Walsh, 2004). Yet, we found that the hemispheres instantiated partially distinct representational geometries that were characterised by specific featural biases. All of these six visual features tested, for example, colour and rectilinearity, were represented in both hemispheres, reflecting clear bilateral processing of visual image statistics. However, the left hemisphere showed preferential coding of rectilinear features, while the right hemisphere was biased toward colour. We found a relative rightward bias for both low and high spatial frequencies. This overall rightward pattern in spatial frequency processing fits with the broader right hemispheric advantage observed across most visual features in this study and aligns with a proposal of a rightward bias in early perceptual processing (Grabowska and Nowicka, 1996). Notably, rectilinearity, for which we found a left hemispheric bias, seemed to reflect later neural processing (onset at 113ms) than the other features (onsets between 76-95 ms), consistent with mid-level processing requiring integration of multiple edge elements and therefore emerging later than simpler features. The biphasic pattern in edge representations, with early right hemisphere dominance followed by a left hemisphere advantage around 150ms, may reflect a transition from coarser global edge processing to finer local contour analysis. Together, these results are consistent with coarse-to-fine theories of visual processing where initial responses reflect broad structure before resolving into finer geometric detail.

Importantly, our results go beyond demonstrating category-level lateralisation, such as the well-established rightward bias for faces and leftward bias for words, to reveal the lower-level feature dimensions that intercorrelate with those asymmetries. Higher-order category preferences in visual cortex have been closely linked to visual feature tuning: for example, face-selective regions coincide with areas that preferentially process curved stimuli and central visual space (Hasson et al., 2002; Yue et al., 2020). The spatial organisation of category selectivity may therefore reflect a map of feature selectivity (Arcaro and Livingstone, 2021, 2024). To assess whether the featural biases observed here were driven by hemispheric asymmetries of specific categories, we ran the same analyses restricted to the 20 objects in the stimulus set. The trend was similar: partial correlations were larger for colour and smaller for rectilinearity in the right than the left hemisphere, suggesting that the featural biases reflect general hemispheric processing differences rather than category-specific effects. The right hemispheric bias for colour we observed thus intercorrelates with the rightward bias for faces, and the left hemispheric bias for rectilinearity intercorrelates with the typical leftward bias for words, though whether these featural biases cause or result from category-level lateralisation remains an open question.

Distinct hemispheric dynamics

The left and right hemispheres exhibited distinct representational dynamics. The left hemisphere showed a smoother temporal profile than the right hemisphere and slower emergence of information for all features except rectilinearity. The right hemisphere, by contrast, showed stronger decoding and more distinct peaks across time, suggesting a more differentiated processing cascade. These different trajectories indicate that the hemispheres are not simply processing the same information at different magnitudes, but are engaged in qualitatively different processing sequences. Object and face representations in particular showed stronger information in the right than left hemisphere, potentially reflecting greater global information in these images and the right hemisphere’s bias towards coarse structure. These representational dynamics suggest that visual information traverses multiple stages of processing within each hemisphere, with each hemisphere contributing different biases on the same input.

Tight coupling and shared representations

Despite hemispheric differences, the time-generalisation analyses revealed that the hemispheres are temporally coupled. The highest across-hemisphere correlations were observed with very short delays, indicating that shared information between the hemispheres is temporally aligned rather than reflecting asymmetric transfer from one hemisphere to the other. This tight coupling suggests rapid interhemispheric communication, consistent with estimates of callosal transfer and our previous findings of interhemispheric transfer using lateralised stimuli (Saron and Davidson, 1989; Robinson et al., 2025). These analyses also revealed an interesting temporal pattern: early processing stages showed limited generalisation across time, consistent with rapidly evolving feedforward representations, while later stages showed broader temporal generalisation. This broadening could reflect sustained and stable high-level processes (King and Dehaene, 2014) and/or higher trial-to-trial temporal variability at later stages of processing (Vidaurre et al., 2018). This temporal pattern was observed in both hemispheres, indicating that both hemispheres follow a similar trajectory despite their different featural biases.

Yet despite this tight coupling, each hemisphere retained unique representational structure not fully captured by the other. Within-hemisphere consistency was higher than across-hemisphere consistency, demonstrating differing representational geometry in keeping with the featural biases observed. However, across-hemisphere correlations were reliably above chance, indicating a degree of shared structure, whether by interhemispheric transfer or independent but similar processing. Representational overlap between the hemispheres points to a degree of redundancy in the system. This redundancy may serve an important functional role, providing robustness to the visual system while the unique information in each hemisphere allows for complementary processing. The distinct yet coupled representations may reflect a fundamental principle by which the brain maximises efficiency and flexibility.

Towards a hemispheric account of visual processing

The present findings complement and extend our previous investigation of hemispheric processing using the same dataset, in which we focused on stimuli presented laterally to one hemifield (Robinson et al., 2025). In that study, contralateral dominance was clear, with stronger and earlier representations in the hemisphere receiving direct visual input. Here, with foveal presentation, both hemispheres received direct early input, yet robust hemispheric asymmetries still emerged, suggesting that these asymmetries reflect intrinsic processing differences rather than being a consequence of contralateral visual field dominance. The tight coupling observed here at near-zero delays contrasts with the transfer delay of approximately 15-30 ms found for lateralised stimuli, suggesting that foveal and peripheral inputs engage different modes of hemispheric coordination; specifically, interhemispheric exchange for foveal stimuli is likely to be more bidirectional than that of peripheral stimulation. Furthermore, while interhemispheric transfer in the lateralised paradigm prioritised conceptual meaning over image statistics (Robinson et al., 2025), here we describe hemispheric biases in visual statistics. Together, these findings suggest a coherent division of labour, such that each hemisphere maintains its own low-level processing signature, and what gets communicated across hemispheres is the higher-level meaning derived from the distinct local computations. These featural asymmetries may therefore reflect each hemisphere's distinct contribution to building a shared high-level representation of the visual world.

The relevance of hemispheric asymmetries extends beyond lateralised paradigms to everyday visual perception, where most critical processing occurs in central vision. The orthogonal target detection task in this experiment suggests that these hemispheric biases reflect inherent lateralisation of visual processing, rather than strategic task differences between the hemispheres. Our stimulus set, spanning objects, faces and words, provided a broad visual space that sampled a wide range of low-level and high-level properties. Although high level visual categories of faces and words have well-documented hemispheric asymmetries (Kanwisher et al., 1997; Cohen et al., 2000), the featural biases were still evident when the analysis was restricted to the object category, ruling out pure category differences driving these effects. Considering multiple stimulus classes together is therefore important for distinguishing general hemispheric processing principles from category-specific effects.

EEG proved to be a valuable tool for investigating hemispheric information processing. Its high temporal resolution allowed us to track the rapid evolution of representations within each hemisphere and to characterise the fine-grained temporal dynamics of interhemispheric similarity. While EEG has limited spatial resolution, the use of lateral clusters combined with multivariate decoding methods is sufficient to reveal robust and distinct representational profiles in each hemisphere (Robinson et al., 2025). Future work using methods with both high temporal and spatial resolution, such as MEG or intracranial recordings, could further refine our understanding of the cortical sources contributing to these hemispheric differences.

In summary, these results demonstrate that the left and right hemispheres make complementary contributions to visual perception, driven by distinct featural biases and different temporal dynamics, while maintaining tight temporal coupling. These findings argue for incorporating hemispheric architecture into models of visual processing, as treating the visual system as a single unified stream overlooks a fundamental organisational principle that shapes how the brain represents the visual world.

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